Slip Face Detection and Route Planning for Small Rovers on Sand Dunes, Tested Against Airborne Lidar

Huda Mueen
Independent Researcher, United Arab Emirates
hudamueen2407@gmail.com

***Abstract*—Small wheeled robots that cross sand dunes usually plan on a global 30 m elevation model, but the main hazard on active dunes, the slip face at the angle of repose, is often metres to tens of metres long. We measure what such maps hide with a preregistered benchmark on six North American dune fields, using one metre airborne lidar as truth and three coarse inputs: 10 m and 30 m block means of the lidar and the real Copernicus GLO-30 product. A subgrid slip face detector trained only on synthetic dune profiles is compared with slope, ruggedness, roughness and interpolation baselines and with gradient boosted trees and a convolutional network trained leave one site out on lidar labels; a released planner is then run with each hazard layer and its routes are scored on face patches in the lidar. At 10 m every method except roughness ranks face cells well (mean AUC 0.93 to 0.97). At 30 m a cubic interpolation of the coarse surface is best or tied best (mean AUC 0.82), the learned models follow, and the synthetic detector (0.74) beats only plain slope; on GLO-30 all methods fall to 0.65 to 0.73. In planning, hazard layers computable from a 30 m map roughly halve the length of route on faces (post hoc) but lift mission success only from 6 % to at most 11 %; at 10 m the best layer reaches 34 %. A post hoc check shows that with a perfect face map, a hard gate on the same 30 m grid still finds a face free route for 83 % of pairs (87 % at 10 m): the main limit is the face information that coarse elevation does not hold.**

***Index Terms*—field robotics, traversability, digital elevation models, sand dunes, slip faces, path planning, benchmark**

I. Introduction

Small wheeled ground robots are increasingly proposed for survey, inspection and search tasks in sand deserts, including the dune seas of the Arabian Peninsula. Before such a robot drives, its route is usually planned on a prior elevation map, and for most deserts the only map available is a global digital elevation model (DEM) with a posting of about 30 m, such as Copernicus GLO-30 [1]. The dominant hazard for a small rover on active dunes is the slip face: the lee side of a dune held near the angle of repose of dry sand, about 30 to 34 degrees [2], where a wheeled vehicle can neither climb nor safely descend or traverse. Slip faces are often only metres to tens of metres long, so a 30 m cell can contain one while its mean slope is gentle.

This paper asks three questions and answers them with preregistered experiments. (i) How much of the slip face hazard does a coarse DEM hide, measured against one metre airborne lidar of real dune fields? (ii) Can a detector that estimates the probability of a hidden face from coarse DEM features recover it, and how does it compare with what a practitioner would otherwise use: slope and roughness costmaps, interpolation of the coarse surface, or a model learned from lidar labels? (iii) Does any of this make planned routes safer when they are scored on the lidar itself?

The contributions are: (1) a benchmark on six North American dune fields with one metre lidar truth and three coarse inputs: 10 m and 30 m block means of the lidar, and the real GLO-30 product, which was acquired years earlier by a different sensor; (2) a subgrid slip face detector trained only on synthetic dune profiles, compared with six baselines including gradient boosted trees and a convolutional network trained leave one site out on lidar labels; (3) an evaluation of a released planner whose hazard layer is swapped between methods, with routes scored on face patches of 10 m^2 or more in the lidar, and perfect label oracles as reference; and (4) a register of predictions written before every run, so that negative results are reported as prominently as positive ones. The answers are sobering: at 30 m no model, ours included, recovers enough of the hazard to make routes face free, the simplest baseline, a cubic interpolation, ranks hidden faces best or tied best, and a post hoc oracle shows that what is missing is mostly face information: with a face map from fine data, a hard gate on the 30 m grid still finds a face free route for most pairs. Code, result files and the prediction register are in the repository listed at the end.

II. Related Work

Traversability from elevation. Surveys of traversability analysis for ground vehicles [3], [4] describe geometric methods that score slope, roughness and step height on a grid, as in the GESTALT navigation software of planetary rovers [5], and in grid map and costmap libraries used with ROS [6], [7]. These methods assume that the grid resolves the hazard. Terrain ruggedness indices [8] are used in geomorphometry for the same purpose, and it has long been known that coarser grids understate slopes [9]. Our slope, ruggedness and roughness baselines follow these definitions on the coarse grid. Risk aware planners propagate terrain uncertainty into routes [10], [11]; our risk weighted policy is a simple member of that family. Learned DEM super resolution [12], [13] would be a natural further baseline; we include only a cubic interpolation.

Slip on sand slopes. Wheel slip on loose sand rises steeply with slope; terramechanics models [14] and online terrain

TABLE I
LIDAR WINDOWS AND THE SHARE OF CELLS THAT HOLD A FACE (CELL LABEL).

Site	Area km^2	Relief m	Face cells 10 m	30 m
Dumont Dunes, CA	14.4	322	32 %	48 %
White Sands, NM	6.3	20	11 %	36 %
Kelso Dunes, CA	9.7	240	11 %	34 %
Great Sand Dunes, CO	14.8	246	13 %	30 %
Algodones Dunes, CA	20.0	122	7 %	24 %
Little Sahara, UT	6.4	99	6 %	31 %

estimation [15] describe it, and field data such as the Dumont Dunes traverses of the Mars Science Laboratory test rover [16] show slip reaching about 50 % at about 13 degrees. Rovers have been immobilised in soft sand on Mars [17], and robophysical studies show how slope, gait and granular flow interact on sandy inclines [18], [19]. Terrain adaptive navigation [20] and learned terrain classification for Mars [10], [21] combine such models with maps; they rely on imagery or DEMs at metre scale.

Prior and overhead maps. Overhead data have been used as priors for ground navigation [22], [23], typically from aerial imagery or lidar at metre scale. The quality of global 30 m DEMs has been assessed against lidar and laser altimetry [24], but we are not aware of an assessment of what such a DEM hides for robot navigation on dunes, or of a benchmark that scores coarse map detectors and planners for this hazard against lidar truth.

III. BENCHMARK

A. Sites and truth

Six windows of USGS 3DEP one metre bare earth lidar [25] cover active dune fields in California, New Mexico, Colorado and Utah (Table I); windows were chosen from 20 m slope images before any metric was computed. Slope is computed with the Horn operator [26]. A post is *steep* at 28 degrees or more. The cell label of a coarse cell is positive when at least 10 steep posts (10 m^2) fall inside its footprint; a secondary label counts only posts between 28 and 36 degrees, to exclude rock and lidar artefacts steeper than sand can stand. For planning, a *face patch* is an 8 connected group of at least 10 steep posts. The two targets differ: a cell can hold part of a patch without holding 10 steep posts, so a perfect detector of the cell label does not remove every face from a route.

B. Inputs

I10 and **I30** are block means of the lidar at 10 and 30 m. **G30** is Copernicus GLO-30 interpolated bilinearly at the centres of the I30 cells. GLO-30 is a surface model derived from TanDEM-X radar acquisitions between December 2010 and January 2015 [1], while the lidar was flown later, so G30 measures what a rover can learn today from a global product, including dune migration and sensor differences.

TABLE II
METHODS COMPARED. LIDAR: TRAINED ON LIDAR LABELS OF THE OTHER FIVE SITES. ALL SCORES ARE PER COARSE CELL; METHODS WITHOUT A PROBABILITY GET A LEAVE ONE SITE OUT LOGISTIC CALIBRATION (SUFFIX -CAL) WHEN ONE IS NEEDED.

Method	Score of a cell	Lidar
DET (ours)	Logistic model on slope, convexity, slope break and local relief, trained on 60 000 synthetic straight crested dune profiles per posting	no
DET-cal	DET recalibrated by logistic regression [28]	yes
SLOPE	Coarse Horn slope, the input of a slope costmap	no
TRI	Terrain ruggedness index [8]	no
ROUGH	RMS residual of a plane fitted to the 3×3 cells, as in GESTALT [5]	no
BICUBIC	Steepest slope after cubic spline interpolation to 1 m	no
GBM	Gradient boosted trees [29], [30] on 14 terrain features (slope, ruggedness, roughness, curvature, topographic position, relief, DET features)	yes
CNN	Fully convolutional network [31], four 3×3 layers dilated by 1, 2, 4 and 8 [32]	yes

C. Methods compared

Table II lists the methods. **DET** (ours) is the subgrid detector of the released planner: its synthetic profiles have stoss slopes of 6 to 20 degrees, faces of 29.5 to 32.5 degrees and heights of 5 to 120 m, are sampled as a DEM would sample them, and the detector never sees lidar. The baselines are what a practitioner would use instead: the input of a slope costmap, two roughness measures from geomorphometry and planetary navigation, an interpolation of the coarse surface, and two models trained leave one site out [27] on lidar labels, which use information DET does not have. No hyperparameter was tuned.

D. Planner and scoring

The released planner assesses a DEM with a friction circle slip model whose four parameters (friction angle, motion resistance, slip scale and planning slip limit) are drawn 128 times inside an assumed range for a small wheeled robot (friction angle 28 to 38 degrees, motion resistance coefficient 0.05 to 0.25, slip scale 0.05 to 0.40, slip limit 0.40 to 0.70; nominal 33 degrees, 0.12, 0.20 and 0.50, which give a climbing limit of 25.5 degrees; this is optimistic against the Dumont data; an exploratory rerun with a posterior fitted to published slip curves, limit 15.9 degrees, routes fewer pairs and changes no conclusion); a cell is passable for a draw when at least four of its eight headings stay below the slip limit. A cell is TRAVERSABLE when it is passable in at least 95 % of draws and its face probability is below 0.20, BLOCKED when passable in at most 5 %, and UNKNOWN otherwise. Routes are least cost paths [33] on the 8 connected grid with cost equal to distance over one minus slip. The *naive* policy uses the slip model only; the *conservative* policy uses TRAVERSABLE cells only; the *risk weighted* policy avoids BLOCKED cells and multiplies the cost of entering a cell by $1 + 5p$, where p is its face probability. Both cautious policies refuse descents of 28 degrees or more between cells. To compare hazard layers, the face probability of the assessment is replaced by each method's probability and the map is reclassified with the planner's own rule; **ORACLE** uses the lidar cell label itself,

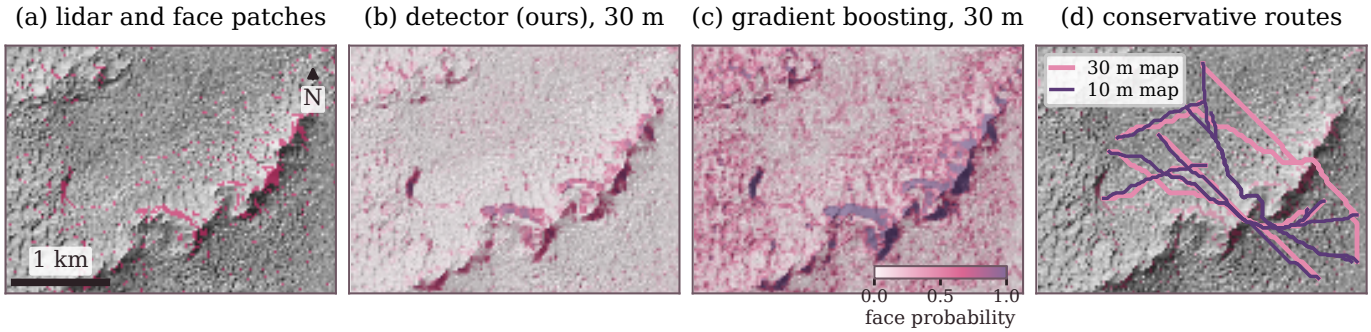


Fig. 1. Kelso Dunes. (a) Hillshade of the one metre lidar with face patches (pink). (b) Face probability of the detector and (c) of gradient boosting on the 30 m block mean. (d) Conservative routes with the detector on the 30 m map (light) and the 10 m map (dark) over the face patches; the 30 m routes cross faces the map cannot see.

and **ORACLE-5** plans on a 5 m map with every cell that holds a face patch post marked as a face.

One hundred origin and destination pairs per site are drawn from the lidar only: centres of 30 m cells whose one metre posts are all gentle, at least 1 km apart. Every method plans between the same points. A route is the polyline through its cell centres, sampled every 0.5 m on the lidar; it is *unsafe* if any sample falls on a face patch, and a *mission success* if a route exists and is not unsafe.

E. Statistics and preregistration

Areas under the ROC curve (AUC) [34] carry 95 % intervals from a spatial block bootstrap [35] with 300 m blocks and 1000 resamples; paired differences use the same resampled blocks. Route outcomes are pooled over sites with intervals from a bootstrap of pairs within sites; paired methods are compared with exact McNemar tests [36] with Holm correction [37], which treat pairs as independent, and, post hoc, with a two stage bootstrap that resamples sites and then pairs. The design, the predictions and their tolerances were committed before any metric was computed (stages 25 and 26 of the register); the baseline code was written in the minutes that followed, and the detection run started shortly before its script was committed (no hash of the script as run was recorded). The detector had been tested on block mean surfaces of the same windows before (stages 12 and 20), which informed the predictions about it, and the planner's descent rule and NO DATA class were added after earlier route tests on these windows. Analyses decided after the runs are labelled post hoc.

IV. Results

A. Detection

Table III and Fig. 2 summarise the six sites. On 10 m block means the hazard is largely visible: every method except plane roughness reaches a mean AUC of at least 0.93, and the detector reaches 0.95 (range 0.91 to 0.97) without any lidar label. The cubic interpolation and the two learned models are nevertheless slightly better than the detector at every site (mean AUC 0.97, 0.97 and 0.97; paired intervals above zero at 6, 6 and 6 of six sites).

TABLE III
Detection: mean over six sites of the AUC and average precision (AP); *trained on lidar labels of the other five sites. Best AUC per input in bold.

	I10		I30		G30	
Method	AUC	AP	AUC	AP	AUC	AP
DET (ours)	0.95	0.76	0.74	0.65	0.65	0.51
SLOPE	0.93	0.73	0.70	0.57	0.65	0.51
TRI	0.95	0.76	0.76	0.64	0.68	0.53
ROUGH	0.77	0.34	0.76	0.60	0.68	0.50
BICUBIC	0.97	0.84	**0.82**	0.72	0.70	0.55
GBM*	**0.97**	0.84	0.81	0.70	0.72	0.57
CNN*	0.97	0.82	0.79	0.69	**0.73**	0.59

On 30 m block means every method loses most of its skill. The best mean AUC, 0.82, belongs to the steepest slope of a cubic interpolation of the coarse surface, a baseline with no training (post hoc paired intervals: better than gradient boosting at 4 sites and tied at the others; better than the network at 4 and worse at 1); the gradient boosted trees and the network, trained on lidar of the other five fields, follow at 0.81 and 0.79. The detector reaches 0.74 (0.64 to 0.86): above the plain slope (0.70), but below the interpolation and the learned models at every site (significant at 6, 6 and 5 of six sites against the interpolation, gradient boosting and the network). The detector also underestimates how common faces are: at its threshold of 0.20 it misses 61 % of face cells on average, and a recalibration on the other sites lowers its Brier score from 0.22 to 0.18 but cannot change its ranking.

On the real Copernicus product every method falls again (mean AUC 0.65 for the detector, 0.70 for the interpolation, 0.72 and 0.73 for the learned models). A post hoc search for the horizontal shift that best aligns GLO-30 with the lidar gives 0 m at Dumont and 0 m at Kelso, where the georeferencing therefore agrees, and 15 m at Algodones and 31 m at White Sands, an actively migrating gypsum dune field [38] whose lidar is years younger than the radar acquisition. There the unsupervised scores are at chance (0.46 to 0.52), and only the learned models retain some skill (0.65 and 0.68).

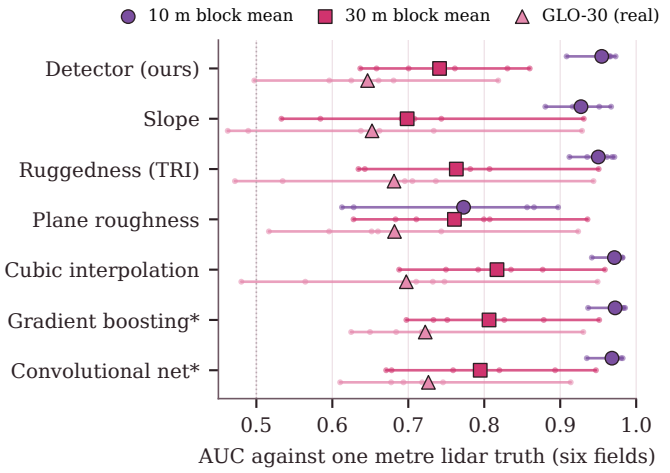


Fig. 2. Face cell detection against one metre lidar on six dune fields. Large markers: mean AUC over sites; small dots and lines: individual sites and their range. *Trained leave one site out on lidar labels.

B. Planning

Table IV and Fig. 3 give the registered outcomes of the 600 pairs. The dune fields are hazardous: the shortest grid path touches a face patch on 96 % of pairs, and the slip model alone, which avoids grades it cannot climb, still touches one on 94 % of pairs at 30 m and 85 % at 10 m. A refusal to route counts against mission success; on a 30 m map that refusal is what the planner's resolution gate intends, so Table IV also gives the unsafe share of the routes that were planned.

On 30 m maps no computable hazard layer produces many face free routes. Mission success is 6 % with the slip model alone and at most 11 % with any computable layer (interpolation, risk weighted: +4.8 points, 95 % interval +2.5 to +7.3 from a bootstrap over sites and pairs, positive at all six sites; no other 30 m layer has an interval entirely above zero). The gating policy fails in two opposite ways. The detector underestimates face prevalence, so it gates few cells: 91 % of pairs are routed and 92 % of those routes are unsafe. The learned and recalibrated layers flag so much of the field that most pairs cannot be routed (19 % for gradient boosting). The layers do reduce exposure (Table V, post hoc): the length of route on a face patch falls from 50 m per km with the slip model alone to 19 m per km with interpolation, gradient boosting or network weighting, and the registered count of face patches entered per km from 5.1 to 2.3, but almost every route still touches some face (no intervals are given for these exposure values). GLO-30 gives the same picture.

On 10 m maps every layer helps. Mission success rises from 15 % with the slip model alone to 24 % with the detector gate, 27 % with the detector weighting and 34 % with gradient boosting weighting, the best computable layer under the registered outcome (+19.2 points over the slip model, interval +10.8 to +26.5, positive at all six sites).

The registered oracle, the stage 25 cell label, reaches 46 % at 10 m and 59 % at 30 m, and the registered 5 m reference, which marks every cell holding a face patch post, reaches

TABLE IV
PLANNING, POOLED OVER SIX SITES (600 PAIRS): UNSAFE SHARE OF ROUTED PAIRS (U) AND MISSION SUCCESS (S), IN %. GATE: CONSERVATIVE POLICY; WEIGHT: RISK WEIGHTED POLICY.

	I10		I30		G30	
Planner	U	S	U	S	U	S
Shortest path	96	4	96	4	96	4
Slip model only	85	15	94	6	95	5
Gate: SLOPE-cal	79	17	88	3	98	0
Gate: BICUBIC-cal	75	22	79	8	91	2
Gate: DET (ours)	72	24	92	7	94	5
Gate: DET-cal	78	22	71	2	75	0
Gate: GBM*	76	24	67	6	70	3
Gate: CNN*	70	27	56	7	79	3
Weight: DET (ours)	73	27	93	7	94	6
Weight: BICUBIC-cal	69	31	89	11	93	7
Weight: GBM*	66	34	93	7	94	6
Gate: ORACLE	54	46	32	59	34	57
ORACLE-5 (5 m)	U 0, S 100					

TABLE V
MISSION SUCCESS (%) WHEN FACE PATCHES OF AT LEAST 10 OR 500 m^2 COUNT (S_{10}; S_{500}, POST HOC), FACE PATCHES ENTERED PER KM (P, REGISTERED SECONDARY OUTCOME) AND METRES OF ROUTE ON A FACE PATCH PER KM (E, POST HOC). FACE MAP (POST HOC): EVERY CELL THAT HOLDS A FACE PATCH POST IS MARKED; WITH THE GATE, NO ROUTE CAN TOUCH A FACE BY CONSTRUCTION.

	I10				I30			
Planner	S_{10}	S_{500}	P	E	S_{10}	S_{500}	P	E
Slip model only	15	40	3.3	24	6	34	5.1	50
Weight: BICUBIC-cal	31	72	1.1	4	11	47	2.3	19
Weight: GBM*	34	67	1.1	5	7	47	2.6	19
Weight: CNN*	29	65	1.2	6	8	48	2.6	19
Gate: ORACLE (cell label)	46	74	0.6	2	59	80	0.1	0.3
Gate: face map	87	87	0.0	0.0	83	83	0.0	0.0
Weight: face map	79	81	0.3	3	55	64	0.6	9

100 %. Most of that gap comes from the label rule, not from the grid (post hoc, Table V): with the same rule on the coarse grids, the conservative policy finds a route for 87 % of pairs at 10 m and 83 % at 30 m. None of these routes can touch a face, by construction, so the informative number is how many pairs stay routable: a face map from fine data lets even a 30 m grid serve most pairs, and the finer grids add the remaining 13 to 17 points. The same face map with the risk weighted policy, which avoids faces only through cost, gives 55 % at 30 m and 79 % at 10 m, so the gain needs a hard gate. When only patches of at least 500 m^2 count (post hoc), every success rate rises (slip model alone 34 % at 30 m and 40 % at 10 m; computable layers up to 48 % and 72 %); the order among the better computable layers changes, but every one stays well below the face map.

C. Preregistered predictions

Table VI lists every registered prediction with its outcome: twelve were met and five were missed. The misses that matter are those against our own method and expectations: interpolation was expected to add nothing over slope at 30 m

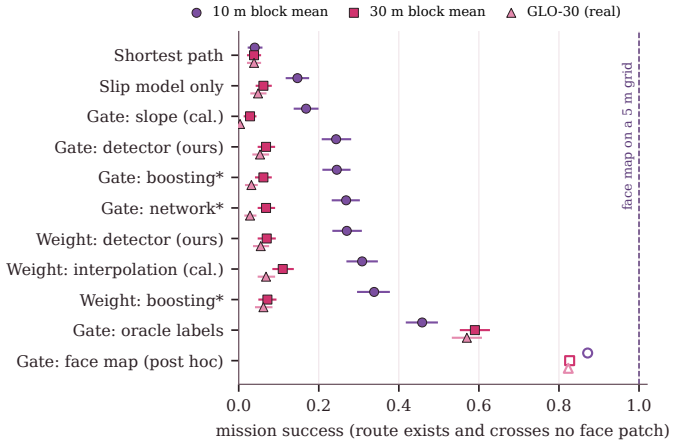


Fig. 3. Mission success of the released planner with swapped hazard layers, pooled over 600 origin and destination pairs (95 % bootstrap intervals). Hollow markers (post hoc): every cell that holds a face patch post marked as a face, on the 10 m and 30 m grids; dashed line: the same on a 5 m grid. *Trained leave one site out on lidar labels.

TABLE VI
THE 17 PREDICTIONS REGISTERED BEFORE THE RUNS (STAGE 25, B; STAGE 26, P), WHAT WAS OBSERVED, AND THE OUTCOME (H HIT, M MISSED).

ID	Prediction	Observed	
B25-1	I30: GBM beats DET at $\geq$3 sites, gain 0 to 0.12	+0.07, 6 sites	H
B25-2	I30: DET above SLOPE at $\geq$4 sites	4 sites	H
B25-3	I30: AUC $\geq$0.90 at $\leq$2 sites, any method	$\leq$1 site	H
B25-4	I10: DET, GBM, CNN $\geq$0.93 everywhere	DET 0.91	M
B25-5	I30: BICUBIC beats SLOPE by $\leq$0.02	+0.12	M
B25-6	DET $\geq$0.03 lower on G30 than I30	0.09 lower	H
B25-7	G30: every method below 0.85	best 0.73	H
B25-8	I30: DET-cal Brier below DET at $\geq$5 sites	6 sites	H
B25-9	I30: CNN within 0.05 of GBM	-0.01	H
P26-1	I30: no layer lifts success by >15 points	oracle +53	M
P26-2	I10: DET gate unsafe $\geq$15 points below naive	13 points	M
P26-3	I30: ORACLE unsafe $\geq$20 points below DET	32 %, 92 %	H
P26-4	ORACLE-5 unsafe $\leq$5 %	0 %	H
P26-5	Shortest path unsafe $\geq$60 %	96 %	H
P26-6	G30 DET gate unsafe $\geq$ I30 minus 5	94 %, 92 %	H
P26-7	I30: GBM unsafe $\leq$10 points below DET	67 %, 92 %	M
P26-8	I10: DET gate length $\leq$1.5 times straight	1.17	H

and became the best method, and the detector gate cut the unsafe share at 10 m by fewer points than predicted.

V. Discussion and Limitations

What the results mean. At 10 m the face hazard is mostly visible and the choice of method matters little; at 30 m it is mostly invisible, and no model of the coarse surface, physical or learned, recovers enough of it to plan face free routes, although the better layers halve the length of route on faces. The post hoc oracle locates most of the limit: given a face map made from fine data, a hard gate on the 30 m grid still finds face free routes for most pairs, so the main missing ingredient is face information; the coarse grid costs a further share of routable pairs, and a soft cost alone does not keep routes off faces. At 30 m the cubic interpolation, which adds no information to the coarse surface, ranks face cells as well as or better than models trained on lidar of five other fields; we suspect that its steepest interpolated slope summarises local curvature more sharply than the features we gave the learned models, but we have not tested this. Our synthetic detector, trained on two dimensional straight crested profiles, ranks worse than both and underestimates how common faces are; its remaining value is that it needs no lidar and gives a probability that a recalibration on any surveyed field can correct. For a planner, a 30 m elevation model is a coarse prior that cannot certify ground as traversable, which the released planner already refuses to do.

Limitations. The six fields are North American; the linear, star and compound megadunes of the Rub' al Khali are larger and only Great Sand Dunes holds star dunes, so the numbers transfer to Arabia as an indication only. The truth counts every surface steeper than 28 degrees, including rock and lidar artefacts at Dumont; the secondary label limited to 36 degrees changes little. The registered planning outcome is strict: any contact with a face patch of 10 m^2 counts as unsafe, although some contacts would be short or survivable for a given vehicle, and a real rover would follow the route with a local planner and its own sensors; the post hoc exposure and patch size analyses show that the absolute numbers depend strongly on this choice and that the order among the better computable layers can change, while the gap between them and the face map remains. The vehicle model is an assumption range, not a measured rover, and no route was driven. GLO-30 differs from the lidar in sensor, date and datum, so the G30 results mix dune migration with product error. Six sites limit the leave one site out training of the learned baselines, and no hyperparameter was tuned for any method.

VI. Conclusion

On six lidar dune fields, a 30 m elevation model hides most slip faces, and neither a physically motivated detector nor models trained on lidar recover enough of them to plan face free routes; on the real Copernicus product the loss is larger still. At 10 m every method sees most faces and the best hazard layer more than doubles mission success over the slip model alone. In a post hoc check, a face map built from fine data let a hard gate on the 30 m grid find face free routes for most pairs, so the main constraint for small rovers on dunes is face information, which global elevation models do not hold. The next step is to run the same benchmark on Arabian dune fields with a drone survey and with a rover whose slip has been measured on the same sand.

Data and Code

USGS 3DEP lidar is in the public domain; Copernicus GLO-30 is used under its licence (produced using Copernicus WorldDEM-30 © DLR e.V. 2010-2014 and © Airbus Defence and Space GmbH 2014-2018, provided under COPERNICUS by the European Union and ESA; all rights reserved). The code, the result files, the prediction register and the scripts that regenerate every number and figure of this paper are at https://github.com/hudasol/dune-rover (stages 25 and 26).

REFERENCES

[1] European Space Agency and Airbus, "Copernicus DEM," Dataset, European Space Agency, 2022.

[2] K. Pye and H. Tsoar, *Aeolian Sand and Sand Dunes*. Berlin, Heidelberg: Springer Berlin Heidelberg, 2009.

[3] P. Papadakis, "Terrain traversability analysis methods for unmanned ground vehicles: A survey," *Engineering Applications of Artificial Intelligence*, vol. 26, no. 4, pp. 1373–1385, 2013.

[4] P. V. K. Borges, T. Peynot, S. Liang, B. Arain, M. Wildie, M. G. Minareci, S. Lichman, G. Samvedi, I. Sa, N. Hudson, M. Milford, P. Moghadam, and P. Corke, "A survey on terrain traversability analysis for autonomous ground vehicles: Methods, sensors, and challenges," *Field Robotics*, vol. 2, pp. 1567–1627, 2022.

[5] S. B. Goldberg, M. W. Maimone, and L. Matthies, "Stereo vision and rover navigation software for planetary exploration," in *Proceedings, IEEE Aerospace Conference*, vol. 5, Big Sky, MT, USA, 2002, pp. 5–2025–5–2036.

[6] P. Fankhauser and M. Hutter, "A universal grid map library: Implementation and use case for rough terrain navigation," in *Robot Operating System (ROS): The Complete Reference (Volume 1)*, ser. Studies in Computational Intelligence, A. Koubaa, Ed. Cham: Springer International Publishing, 2016, vol. 625, pp. 99–120.

[7] S. Macenski, F. Martin, R. White, and J. Ginés Clavero, "The Marathon 2: A navigation system," in *2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2020, pp. 2718–2725.

[8] S. J. Riley, S. D. DeGloria, and R. Elliot, "A terrain ruggedness index that quantifies topographic heterogeneity," *Intermountain Journal of Sciences*, vol. 5, no. 1-4, pp. 23–27, 1999. [Online]. Available: https://arc.lib.montana.edu/ojs/index.php/IJS/article/view/2914

[9] W. Zhang and D. R. Montgomery, "Digital elevation model grid size, landscape representation, and hydrologic simulations," *Water Resources Research*, vol. 30, no. 4, pp. 1019–1028, 1994.

[10] M. Ono, T. J. Fuchs, A. Steffy, M. Maimone, and J. Yen, "Risk-aware planetary rover operation: Autonomous terrain classification and path planning," in *2015 IEEE Aerospace Conference*, Big Sky, MT, 2015, pp. 1–10.

[11] D. D. Fan, K. Otsu, Y. Kubo, A. Dixit, J. Burdick, and A.-a. Aghamohammadi, "STEP: Stochastic traversability evaluation and planning for risk-aware off-road navigation," in *Robotics: Science and Systems XVII*, 2021.

[12] Z. Chen, X. Wang, Z. Xu, and W. Hou, "Convolutional neural network based DEM super resolution," in *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, vol. XLI-B3, 2016, pp. 247–250.

[13] Z. Xu, Z. Chen, W. Yi, Q. Gui, W. Hou, and M. Ding, "Deep gradient prior network for DEM super-resolution: Transfer learning from image to DEM," *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 150, pp. 80–90, 2019.

[14] G. Ishigami, A. Miwa, K. Nagatani, and K. Yoshida, "Terramechanics-based model for steering maneuver of planetary exploration rovers on loose soil," *Journal of Field Robotics*, vol. 24, no. 3, pp. 233–250, 2007.

[15] K. Iagnemma, S. Kang, H. Shibly, and S. Dubowsky, "Online terrain parameter estimation for wheeled mobile robots with application to planetary rovers," *IEEE Transactions on Robotics*, vol. 20, no. 5, pp. 921–927, 2004.

[16] M. Heverly, J. Matthews, J. Lin, D. Fuller, M. Maimone, J. Biesiadecki, and J. Leichty, "Traverse performance characterization for the Mars Science Laboratory rover," *Journal of Field Robotics*, vol. 30, no. 6, pp. 835–846, 2013.

[17] R. E. Arvidson, J. F. Bell, P. Bellutta, N. A. Cabrol, J. G. Catalano, J. Cohen, L. S. Crumpler, D. J. Des Marais *et al.*, "Spirit Mars rover mission: Overview and selected results from the northern Home Plate Winter Haven to the side of Scamander crater," *Journal of Geophysical Research: Planets*, vol. 115, no. E7, 2010.

[18] H. Marvi, C. Gong, N. Gravish, H. Astley, M. Travers, R. L. Hatton, J. R. Mendelson, H. Choset, D. L. Hu, and D. I. Goldman, "Sidewinding with minimal slip: Snake and robot ascent of sandy slopes," *Science*, vol. 346, no. 6206, pp. 224–229, 2014.

[19] S. Shrivastava, A. Karsai, Y. O. Aydin, R. Pettinger, W. Bluethmann, R. O. Ambrose, and D. I. Goldman, "Material remodeling and unconventional gaits facilitate locomotion of a robophysical rover over granular terrain," *Science Robotics*, vol. 5, no. 42, p. eaba3499, 2020.

[20] D. Helmick, A. Angelova, and L. Matthies, "Terrain adaptive navigation for planetary rovers," *Journal of Field Robotics*, vol. 26, no. 4, pp. 391–410, 2009.

[21] B. Rothrock, R. Kennedy, C. Cunningham, J. Papon, M. Heverly, and M. Ono, "SPOC: Deep learning-based terrain classification for Mars rover missions," in *AIAA SPACE 2016*. Long Beach, California: American Institute of Aeronautics and Astronautics, 2016.

[22] B. Sofman, E. Lin, J. A. Bagnell, J. Cole, N. Vandapel, and A. Stentz, "Improving robot navigation through self-supervised online learning," *Journal of Field Robotics*, vol. 23, no. 11-12, pp. 1059–1075, 2006.

[23] D. Silver, J. A. Bagnell, and A. Stentz, "Learning from demonstration for autonomous navigation in complex unstructured terrain," *The International Journal of Robotics Research*, vol. 29, no. 12, pp. 1565–1592, 2010.

[24] P. L. Guth and T. M. Geoffroy, "LiDAR point cloud and ICESat-2 evaluation of 1 second global digital elevation models: Copernicus wins," *Transactions in GIS*, vol. 25, no. 5, pp. 2245–2261, 2021.

[25] L. J. Sugarbaker, E. W. Constance, H. K. Heidemann, A. L. Jason, V. Lukas, D. L. Saghy, and J. M. Stoker, "The 3D Elevation Program initiative: A call for action," U.S. Geological Survey, Circular 1399, 2014.

[26] B. K. P. Horn, "Hill shading and the reflectance map," *Proceedings of the IEEE*, vol. 69, no. 1, pp. 14–47, 1981.

[27] D. R. Roberts, V. Bahn, S. Ciuti, M. S. Boyce, J. Elith, G. Guillera-Arroita, S. Hauenstein, J. J. Lahoz-Monfort, B. Schröder, W. Thuiller, D. I. Warton, B. A. Wintle, F. Hartig, and C. F. Dormann, "Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure," *Ecography*, vol. 40, no. 8, pp. 913–929, 2017.

[28] J. C. Platt, "Probabilities for SV machines," in *Advances in Large-Margin Classifiers*, A. J. Smola, P. Bartlett, B. Schölkopf, and D. Schuurmans, Eds. Cambridge, MA: The MIT Press, 2000, pp. 61–74.

[29] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, "LightGBM: A highly efficient gradient boosting decision tree," in *Advances in Neural Information Processing Systems*, vol. 30, 2017. [Online]. Available: https://proceedings.neurips.cc/paper/2017/hash/6449f44a102fde848669bdd9eb6b76fa-Abstract.html

[30] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and É. Duchesnay, "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, no. 85, pp. 2825–2830, 2011. [Online]. Available: https://jmlr.org/papers/v12/pedregosa11a.html

[31] A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimelshein, L. Antiga, A. Desmaison, A. Kopf, E. Yang, Z. DeVito, M. Raison, A. Tejani, S. Chilamkurthy, B. Steiner, L. Fang, J. Bai, and S. Chintala, "PyTorch: An imperative style, high-performance deep learning library," in *Advances in Neural Information Processing Systems*, vol. 32, 2019. [Online]. Available: https://proceedings.neurips.cc/paper/2019/hash/bdbca288fee7f92f2bfa9f7012727740-Abstract.html

[32] F. Yu and V. Koltun, "Multi-scale context aggregation by dilated convolutions," in *International Conference on Learning Representations (ICLR)*, 2016. [Online]. Available: https://arxiv.org/abs/1511.07122

[33] E. W. Dijkstra, "A note on two problems in connexion with graphs," *Numerische Mathematik*, vol. 1, no. 1, pp. 269–271, 1959.

[34] J. A. Hanley and B. J. McNeil, "The meaning and use of the area under a receiver operating characteristic (ROC) curve," *Radiology*, vol. 143, no. 1, pp. 29–36, 1982.

[35] H. R. Künsch, "The jackknife and the bootstrap for general stationary observations," *The Annals of Statistics*, vol. 17, no. 3, pp. 1217–1241, 1989.

[36] Q. McNemar, "Note on the sampling error of the difference between correlated proportions or percentages," *Psychometrika*, vol. 12, no. 2, pp. 153–157, 1947.

[37] S. Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, pp. 65–70, 1979. [Online]. Available: https://zbmath.org/3624650

[38] A. Pedersen, G. Kocurek, D. Mohrig, and V. Smith, "Dune deformation in a multi-directional wind regime: White Sands Dune Field, New Mexico," *Earth Surface Processes and Landforms*, vol. 40, no. 7, pp. 925–941, 2015.